Gap Orientation and Binomial Coefficients Modulo Products of Three Nearby Primes

An affirmative answer to Erdős Problem #700(b) of Erdős and Szekeres

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Abstract

For a composite integer n let $f(n) = \min_{1<k\le n/2} \gcd\big(n, \binom{n}{k}\big)$. In 1978 Erdős and Szekeres asked whether $f(n) > n^{1/2}$ for infinitely many composite n. They could not prove it, and the question (Erdős Problem #700(b)) is listed as open. We answer it affirmatively. The main tool is an exact evaluation of $f(pqr)$ for primes $p < q < r$ that are close together. If $2(r-p)^2 < p$, then

$$f(pqr) = pq \ \text{ if } r-q > q-p, \qquad f(pqr) = p \ \text{ if } r-q \le q-p.$$

So in this range $f(pqr)$ exceeds $(pqr)^{1/2}$ exactly when the gaps between the prime factors increase. To prove this we write each product of two of the primes in the base of the third. The digits are simple explicit expressions in the two gaps, and whether the middle digit of pr in base q carries a borrow depends only on the sign of $(r-q)-(q-p)$. Lucas' theorem then turns the divisibility conditions into linear inequalities. A counting lemma for triples with increasing gaps, combined with Chebyshev's bound and Jensen's inequality, then gives $\gg N^{1/2}/(\log N)^2$ composite $n \le N$ with $f(n) > n^{1/2}$. All of them satisfy $f(n) = n/P(n) > (n/2)^{2/3}$. We check the evaluation by direct computation of f for all 102,122 admissible triples with $p \le 30{,}000$, and we spot-check it with an independent computation based on Legendre's formula. We also show that the constant 2 cannot be lowered to 1.75.

1 Introduction

For a composite integer n put

$$f(n) = \min_{1<k\le n/2} \gcd\Big(n, \binom{n}{k}\Big).$$

This function was studied by Erdős and Szekeres [1]. Two simple facts frame their questions. First, $\binom{n}{k}$ is divisible by $n/\gcd(n,k)$ (Lemma 2.3), so

$$f(n) \ \ge \ p(n), \tag{1}$$

where $p(n)$ denotes the least prime factor of n. Second, if n is not a prime power and $s^e \,\|\, n$ with s prime, then $\gcd(n, \binom{n}{s^e}) = n/s^e$. Hence $f(n) \le n/Q(n) \le n/P(n)$, where $Q(n)$ is the largest prime power dividing n and $P(n)$ the largest prime factor of n. Erdős and Szekeres proved $f(n) \le (1+o(1))\, n/\log n$. They noted that $f(n) = n/P(n)$ when n is a product of two primes, and also for $n = 30$. The problem collection [2] records three questions (Erdős Problem #700):

(a) characterise the composite n with $f(n) = n/P(n)$;[1]

[1] In [1] this question is stated with the greatest prime power dividing n, that is, with $n/Q(n)$; the problem page [2] uses the largest prime factor. For the squarefree n studied here $Q(n) = P(n)$, so our results apply to both readings.

(b) are there infinitely many composite n such that $f(n) > n^{1/2}$?

(c) is it true that $f(n) \ll_A n/(\log n)^A$ for every $A > 0$?

Erdős and Szekeres could not prove (b). For every prime s we have $f(s^2) = s$: the lower bound is (1), and $k = s$ gives $\gcd(s^2, \binom{s^2}{s}) = s$ because adding s and $s^2 - s$ in base s produces exactly one carry. So the inequality in (b) must be strict for the question to have content. Problem #700 is listed as open on the problem page [2] and in the community database [3], which records one status for the whole problem ("open", last updated 31 August 2025, in the data of 28 October 2026). The questions are formalised in Lean in [4]; see also Guy [5, B31, B33], and OEIS A091963 [6], which [3] lists as possibly related.

We answer (b) affirmatively. The construction is explicit: products of three primes that lie close together, with increasing gaps.

Theorem 1.1 (gap orientation)**.** *Let $p < q < r$ be primes with*

$$2(r-p)^2 < p, \tag{2}$$

and let $n = pqr$. Then

$$f(n) = \begin{cases} pq & \text{if } r - q > q - p, \\ p & \text{if } r - q \le q - p. \end{cases}$$

In particular $f(n) > n^{1/2}$ if and only if $r - q > q - p$, and in that case $f(n) = n/P(n) > (n/2)^{2/3}$.

So within the range (2) the value $f(pqr)$ is either the smallest value $p(n)$ allowed by (1) or the largest value $n/P(n)$ allowed by the upper bound. The orientation of the two gaps decides which one occurs.

Theorem 1.2 (answer to Erdős Problem #700(b))**.** *There are infinitely many composite n with $f(n) > n^{1/2}$. More precisely, there are an absolute constant $c > 0$ and an N_0 such that*

$$\#\{\, n \le N : \ n \text{ composite}, \ f(n) > n^{1/2} \,\} \ \ge \ c\,\frac{N^{1/2}}{(\log N)^2} \qquad (N \ge N_0).$$

Every integer produced by the proof has exactly three prime factors and satisfies $f(n) = n/P(n) > (n/2)^{2/3}$.

Corollary 1.3 (towards (a))**.** *There are infinitely many composite n with exactly three prime factors and $f(n) = n/P(n)$. For $n = pqr$ with $p < q < r$ satisfying (2), the equality $f(n) = n/P(n)$ holds if and only if $r - q > q - p$.*

Erdős and Szekeres gave $n = 30$ as an example of $f(n) = n/P(n)$ beyond the products of two primes. Corollary 1.3 supplies an infinite family with an exact criterion. A full characterisation, as asked in (a), remains open (Section 7).

The problem, the method, and the application. The proof follows three steps.

A problem. Call a squarefree $n = pqr$ with $p < q < r$ and $pq > r$ *two-witness rigid* if every $\binom{n}{k}$ with $1 < k \le n/2$ is divisible by at least two of p, q, r. Lemma 2.4 shows that $f(n) > n^{1/2}$ is equivalent to rigidity. By Lucas' theorem, rigidity fails exactly when some multiple $k = st\,j$ of two of the primes is digit-dominated by n in both bases s and t. Which triples are rigid?

A method: gap coordinates and the borrow switch. Write $q = p + a$ and $r = q + c$. When a and c are small, each product of two of the primes, written in the base of the third, has digits given by simple expressions in a, c and the base (Lemma 3.1):

$$qr = \langle 1,\ 2a + c,\ a(a + c)\rangle_p, \qquad pq = \langle r - a - 2c,\ c(a + c)\rangle_r,$$

$$pr = \begin{cases} \langle 1,\ c-a-1,\ q-ac\rangle_q & (c > a), \\ \langle q-(a-c)-1,\ q-ac\rangle_q & (c \le a). \end{cases}$$

The sign of $c-a$ decides whether a borrow passes through the middle digit of pr in base q. With these digits each simultaneous domination becomes a pair of linear inequalities in the multiplier j. They are contradictory when $c > a$. When $c \le a$ they have an explicit solution.

The application. A counting lemma for triples with increasing gaps in a short interval (Lemma 5.1) needs nothing about primes beyond their number. Combined with Chebyshev's bound and Jensen's inequality, it gives many admissible triples of primes, and Theorem 1.2 follows.

All three ingredients rest on equivalent forms of one classical fact: Lucas' digit criterion, Kummer's carry count and the congruence $(1+X)^n \equiv (1+X^s)^{n/s} \pmod{s}$. The new ingredient is the pair of explicit gap-coordinate expansions and the observation that a single sign governs f.

Computations. Section 6 reports exhaustive checks of Theorem 1.1, a check by Legendre's formula that does not use Lucas' theorem, tests showing that (2) is nearly sharp, and data on f for small n. The programs are reproduced in the appendix.

2 Preliminaries

For a prime s and integers $x, y \ge 0$ we write $x \preceq_s y$ if every base-s digit of x is at most the corresponding digit of y. We write $\langle d_m, \dots, d_1, d_0\rangle_s = \sum_i d_i s^i$ when $0 \le d_i < s$ for all i.

Lemma 2.1 (Lucas [7]; Kummer [8])**.** *Let s be prime and $0 \le x \le y$. Then $s \nmid \binom{y}{x}$ if and only if $x \preceq_s y$. Equivalently, there is no carry when x and $y-x$ are added in base s.*

Lemma 2.2. *Let n be squarefree and $0 < k < n$.*

(i) $\gcd(n, \binom{n}{k})$ *is the product of the primes $s \mid n$ with $s \mid \binom{n}{k}$.*

(ii) *If $s \mid n$ and $s \nmid \binom{n}{k}$, then $s \mid k$ and $k/s \preceq_s n/s$.*

Proof. (i) is clear because n is squarefree. (ii) The last base-s digit of n is 0. By Lemma 2.1 the last digit of k is 0, so $s \mid k$. Deleting the last digits of k and n leaves k/s and n/s. □

Lemma 2.3. *For composite n and $1 < k \le n/2$, the integer $n/\gcd(n,k) > 1$ divides $\gcd(n, \binom{n}{k})$. Consequently $f(n) \ge p(n)$.*

Proof. From $k\binom{n}{k} = n\binom{n-1}{k-1}$ we get $n \mid k\binom{n}{k}$, hence $n/\gcd(n,k)$ divides $\binom{n}{k}$. It also divides n. It exceeds 1 since $\gcd(n,k) \le k < n$, and every divisor of n exceeding 1 is at least $p(n)$. □

Lemma 2.4 (two-witness reformulation)**.** *Let $n = pqr$ with odd primes $p < q < r$ and $pq > r$.*

(i) *$f(n) > n^{1/2}$ if and only if, for every integer k with $1 < k \le n/2$, at least two of p, q, r divide $\binom{n}{k}$. In that case $f(n) \ge pq$.*

(ii) *Let $\{s,t,u\} = \{p,q,r\}$. If $1 < k \le n/2$ and neither s nor t divides $\binom{n}{k}$, then $k = stj$ for an integer j with $1 \le j \le (u-1)/2$, and*

$$tj \ \preceq_s \ tu \qquad \text{and} \qquad sj \ \preceq_t \ su.$$

Proof. (i) By Lemma 2.2(i), $\gcd(n, \binom{n}{k})$ is the product of those of p, q, r that divide $\binom{n}{k}$. A product of at least two of them is at least $pq > n^{1/2}$, because $pq > r$. A product of at most one of them is at most $r < n^{1/2}$, because $r < pq$. (ii) By Lemma 2.2(ii), $s \mid k$ and $t \mid k$, so $k = st\,j$ with $j \ge 1$. From $k \le n/2 = stu/2$ we get $j \le u/2$, so $j \le (u-1)/2$ because u is odd. Finally $k/s = tj \preceq_s n/s = tu$ and $k/t = sj \preceq_t n/t = su$ by Lemma 2.2(ii). □

3 Gap coordinates

In Sections 3 and 4, $p < q < r$ are primes satisfying (2), and

$$a = q - p, \qquad c = r - q, \qquad b = a + c = r - p.$$

Condition (2) forces $p > 2$, so all three primes are odd. Hence a and c are even, $a, c \geq 2$, $b \geq 4$, and $p > 32$. Since $a, c < b$, condition (2) gives

$$ab,\ ac,\ bc,\ c^2,\ a(a+b),\ b(a+b),\ c(b+c)\ <\ 2b^2\ <\ p, \qquad a+2b < 3b < p, \quad b+2a < p. \tag{3}$$

Every quantity in (3) is therefore also smaller than q and r. We also use $b < \sqrt{p/2} < p$ and $r = p + b < 2p$.

Lemma 3.1 (gap-coordinate digits)**.** *Under* (2)*:*

(E1) $qr = \langle 1,\ a+b,\ ab\rangle_p$*;*

(E2) $pq = \langle r-b-c,\ bc\rangle_r$*;*

(E3) $pr = \langle 1,\ c-a-1,\ q-ac\rangle_q$ *if* $c > a$*, and* $pr = \langle q+c-a-1,\ q-ac\rangle_q$ *if* $c \leq a$*.*

Proof. (E1) $qr = (p+a)(p+b) = p^2 + (a+b)p + ab$, with $0 \leq a+b < p$ and $0 < ab < p$ by (3). (E2) $pq = (r-b)(r-c) = r^2 - (b+c)r + bc = (r-b-c)\,r + bc$, with $0 \leq r-b-c < r$ (as $b+c < 2b < r$) and $0 < bc < r$. (E3) $pr = (q-a)(q+c) = q^2 + (c-a)q - ac$, and $0 < ac < q$. If $c > a$, this equals $q^2 + (c-a-1)q + (q-ac)$ with $0 \leq c-a-1 < q$. If $c \leq a$, it equals $(q+c-a-1)q + (q-ac)$ with $0 \leq q+c-a-1 \leq q-1$, since $1 \leq a-c+1 \leq a < q$. □

The case distinction in (E3) is the *borrow switch*. When $c \leq a$ the coefficient $c-a$ is not positive. A borrow from the leading digit then makes the middle digit almost as large as q, and the corresponding domination condition becomes vacuous.

4 Proof of Theorem 1.1

We treat the three pairs of primes separately.

Lemma 4.1 (the pair $\{p, q\}$)**.** *Assume* $c > a$*. No integer* j *with* $1 \leq j \leq (r-1)/2$ *satisfies both* $qj \preceq_p qr$ *and* $pj \preceq_q pr$*.*

Proof. Suppose that j does. Write $aj = \gamma p + \varepsilon$ with $0 \leq \varepsilon < p$. Then $qj = pj + aj = (j+\gamma)p + \varepsilon$. Since $j \leq (r-1)/2 < (p+b)/2 < p$ (recall $r < 2p$), we have $\gamma \leq aj/p < a$, hence $j + \gamma < (p+b)/2 + a \leq p$ by (3). So the base-p digits of qj are $(j+\gamma, \varepsilon)$, and (E1) forces $j + \gamma \leq a + b$ and $\varepsilon \leq ab$. Hence $j \leq a + b$, so $aj \leq a(a+b) < p$, which gives $\gamma = 0$ and $\varepsilon = aj$. Now $aj \leq ab$ gives $j \leq b$.

Since $0 < aj \leq ab < q$, we have $pj = (q-a)j = (j-1)q + (q-aj)$, that is, $pj = \langle j-1,\ q-aj\rangle_q$. By (E3) with $c > a$, the relation $pj \preceq_q pr$ requires $j - 1 \leq c - a - 1$ and $q - aj \leq q - ac$, that is, $j \leq c - a$ and $j \geq c$. These are incompatible because $a > 0$. □

Lemma 4.2 (the pair $\{p, r\}$)**.** *No integer* j *with* $1 \leq j \leq (q-1)/2$ *satisfies both* $rj \preceq_p rq$ *and* $pj \preceq_r pq$*. Here no assumption on the orientation of the gaps is needed.*

Proof. Suppose that j does. Write $bj = \gamma p + \varepsilon$ with $0 \leq \varepsilon < p$, so $rj = (j+\gamma)p + \varepsilon$. From $j \leq (q-1)/2 < (p+a)/2 < p$ we get $\gamma \leq bj/p < b$, hence $j + \gamma < (p+a)/2 + b \leq p$ by (3). By (E1) (note $rq = qr$) we get $j + \gamma \leq a + b$. Hence $bj \leq b(a+b) < p$, so $\gamma = 0$ and $\varepsilon = bj \leq ab$, that is, $j \leq a$.

Then $0 < bj \leq ab < r$, so $pj = (r-b)j = \langle j-1,\ r-bj\rangle_r$. By (E2), $pj \preceq_r pq$ requires $r - bj \leq bc$. Therefore $r \leq b(j+c) \leq b(a+c) = b^2$, which contradicts $b^2 < p < r$. □

Lemma 4.3 (the pair $\{q, r\}$). *Assume $c > a$. No integer j with $1 \le j \le (p-1)/2$ satisfies both $rj \preceq_q rp$ and $qj \preceq_r qp$.*

Proof. Suppose that j does. Write $cj = \gamma q + \varepsilon$ with $0 \le \varepsilon < q$, so $rj = (q+c)j = (j+\gamma)q + \varepsilon$. Since $j < p/2 < q/2$, we have $\gamma \le cj/q < c/2$, and $j + \gamma < (q+c)/2 < q$. So rj has base-q digits $(j+\gamma, \varepsilon)$. By (E3) with $c > a$, its middle digit must satisfy $j + \gamma \le c - a - 1$. Hence $j \le c - a - 1 < c$.

Then $0 < cj < c^2 < r$, so $qj = (r-c)j = \langle j-1,\ r-cj\rangle_r$. By (E2), $qj \preceq_r qp$ requires $r - cj \le bc$. Therefore $r \le c(j+b) \le c(c-a-1+b) < c(b+c) < p$, a contradiction. □

Lemma 4.4 (an explicit witness when $c \le a$). *Assume $c \le a$, and put $j^* = \lceil (r-bc)/c \rceil$. Then $1 \le j^* \le (p-1)/2$, $rj^* \preceq_q rp$ and $qj^* \preceq_r qp$.*

Proof. Since $bc < r$ we have $j^* \ge 1$. By definition,

$$r - bc \ \le\ cj^* \ \le\ r - bc + c - 1. \tag{4}$$

Range. As $c \ge 2$, $j^* \le (r-bc+c-1)/c = (r-1)/c - b + 1 \le (r-1)/2 - b + 1 = (p-b+1)/2 \le (p-1)/2$.

Base q. We have $q - ac = r - c - ac$. The right side of (4) is at most $r - c - ac$, since this is equivalent to $2c - 1 \le (b-a)c = c^2$, that is, to $(c-1)^2 \ge 0$. Hence $0 < cj^* \le q - ac < q$, and $rj^* = (q+c)j^* = \langle j^*,\ cj^*\rangle_q$. By (E3) with $c \le a$, $rp = \langle q+c-a-1,\ q-ac\rangle_q = \langle p+c-1,\ q-ac\rangle_q$. As $j^* < p + c - 1$ and $cj^* \le q - ac$, we get $rj^* \preceq_q rp$.

Base r. By (4) and $bc \ge c$, we have $0 < r - bc \le cj^* < r$. Hence $qj^* = (r-c)j^* = \langle j^* - 1,\ r - cj^*\rangle_r$ with $0 < r - cj^* \le bc$. By (E2), $qp = \langle r - b - c,\ bc\rangle_r = \langle p - c,\ bc\rangle_r$. Since $j^* - 1 \le (p-3)/2 < p - c$ and $r - cj^* \le bc$, we get $qj^* \preceq_r qp$. □

Proof of Theorem 1.1. Since $r = p + b < 2p \le pq$, Lemma 2.4 applies, and $p < n^{1/2} < pq$.

Case $c > a$. Let $1 < k \le n/2$. Suppose two of p, q, r do not divide $\binom{n}{k}$. Lemma 2.4(ii) then gives an integer j of one of three kinds: for the pair $\{s, t\} = \{p, q\}$ ($u = r$), a $j \le (r-1)/2$ with $qj \preceq_p qr$ and $pj \preceq_q pr$; for $\{p, r\}$ ($u = q$), a $j \le (q-1)/2$ with $rj \preceq_p rq$ and $pj \preceq_r pq$; for $\{q, r\}$ ($u = p$), a $j \le (p-1)/2$ with $rj \preceq_q rp$ and $qj \preceq_r qp$. Lemmas 4.1, 4.2 and 4.3 exclude these three cases. Hence at least two of the primes divide every such $\binom{n}{k}$, and $f(n) \ge pq$ by Lemma 2.4(i).

For the upper bound take $k = r$. Since $p \nmid r$, the last base-p digit of r is non-zero while that of n is 0, so $p \mid \binom{n}{r}$ by Lemma 2.1; likewise $q \mid \binom{n}{r}$. In base r we have $n = \langle r - b - c,\ bc,\ 0\rangle_r$ by (E2), and $r = \langle 1, 0\rangle_r$ with $1 \le bc$, so $r \nmid \binom{n}{r}$. Thus $\gcd(n, \binom{n}{r}) = pq$ and $f(n) = pq$.

Case $c \le a$. Let $k^* = qr\, j^*$ with j^* from Lemma 4.4. Then $1 < k^* \le qr(p-1)/2 < n/2$. The last base-$q$ digits of k^* and n are 0, and $k^*/q = rj^* \preceq_q rp = n/q$, so $k^* \preceq_q n$ and $q \nmid \binom{n}{k^*}$. In the same way $r \nmid \binom{n}{k^*}$. Since $1 \le j^* < p$, $p \nmid k^*$, so $p \mid \binom{n}{k^*}$. Hence $\gcd(n, \binom{n}{k^*}) = p$, and together with (1), $f(n) = p$.

Finally, if $f(n) = pq = n/r$, then $f(n)^3 > n^2/4$ is equivalent to $4pq > r^2$, which holds because $r < 2p$. So $f(n) > (n/2)^{2/3}$. □

Remark 4.5. Each of Lemmas 4.1 and 4.3 uses (2) through a single inequality: $a(a+b) < p$ and $c(b+c) < r$ respectively. Lemma 4.2 uses $b(a+b) < p$ and $b^2 < r$. Section 6 shows that the constant 2 in (2) cannot be replaced by 1.75. The failures found there come from the pair $\{q, r\}$, where $c(b+c) < r$ fails.

5 Many triples with increasing gaps

Lemma 5.1 (counting increasing triples). *Let $L \ge 2$, and let $x_1 < x_2 < \cdots < x_k$ be integers with $x_k - x_1 < L$. Then the number of triples $i < j < l$ with $x_l - x_j > x_j - x_i$ is at least $\frac{1}{2}\big((k - 3 - \log_2 L)^+\big)^2$.*

Proof. Fix $l \in \{3,\dots,k\}$ and put $D_t = x_l - x_t$ for $1 \le t < l$. Call $j \in \{2,\dots,l-1\}$ *bad* if $x_l - x_j \le x_j - x_{j-1}$, that is, if $D_j \le D_{j-1} - D_j$. For every j we have $D_{j-1} > D_j$, and for bad j we have $D_{j-1} \ge 2D_j$. Since $D_{l-1} \ge 1$ and $D_1 < L$,

$$2^{\#\text{bad}} \;\le\; \prod_{j=2}^{l-1} \frac{D_{j-1}}{D_j} = \frac{D_1}{D_{l-1}} < L,$$

so fewer than $\lambda = \log_2 L$ indices are bad. Each good j gives the triple $(j-1, j, l)$, and different pairs (j,l) give different triples. The number of triples is therefore at least

$$\sum_{l=3}^{k} (l - 2 - \lambda)^+ \;\ge\; \sum_{m=\lceil\lambda\rceil}^{k-2} (m - \lceil\lambda\rceil) = \tfrac{1}{2}T(T+1), \qquad T = k - 2 - \lceil\lambda\rceil,$$

and $\frac{1}{2}T(T+1) \ge \frac{1}{2}\big((k-3-\lambda)^+\big)^2$ because $\lceil\lambda\rceil < \lambda + 1$. (If $T < 0$ the bound is 0.) □

The extreme case of Lemma 5.1 is a set whose gaps at least double from right to left, such as $x_t = -2^{k-t}$ $(1 \le t \le k)$. Such a set has no increasing triple at all, while its span is only $2^{k-1} - 1$, so some dependence on $\log_2 L$ is unavoidable.

Proof of Theorem 1.2. Let X be large, $L = \lfloor\sqrt{X/2}\rfloor$ and $M = \lceil X/L\rceil$. The intervals $I_t = [X + tL,\, X + (t+1)L)$, $0 \le t < M$, cover $[X, 2X)$. Let k_t be the number of primes in I_t. By Chebyshev-type bounds we have

$$\sum_t k_t \;\ge\; \pi(2X) - \pi(X) \;\ge\; \frac{X}{2\log X}$$

for $X \ge 150$; explicitly, this follows from the bounds $x/\log x < \pi(x) < 1.25506\,x/\log x$ of Rosser and Schoenfeld [9] (the lower one for $x \ge 17$).

Three primes $p < q < r$ in the same I_t satisfy $r - p \le L - 1 < \sqrt{X/2} \le \sqrt{p/2}$, so (2) holds. By Lemma 5.1, I_t contains at least $\varphi(k_t)$ triples of primes with $r - q > q - p$, where $\varphi(x) = \frac{1}{2}\big((x - 3 - \log_2 L)^+\big)^2$. By Theorem 1.1 each such triple gives $n = pqr$ with $f(n) = pq > n^{1/2}$ and $n < (2X + L)^3 < 9X^3$. Different triples give different n by unique factorisation.

The function φ is convex, so Jensen's inequality gives $\sum_t \varphi(k_t) \ge M\varphi\big(\frac{1}{M}\sum_t k_t\big)$. Now $X/L \le M \le 2X/L$, so $\frac{1}{M}\sum_t k_t \ge L/(4\log X)$. For large X we also have $L/(4\log X) - 3 - \log_2 L \ge L/(5\log X)$. Hence

$$\sum_t \varphi(k_t) \;\ge\; \frac{X}{L}\cdot\frac{1}{2}\Big(\frac{L}{5\log X}\Big)^2 = \frac{XL}{50(\log X)^2} \;\ge\; \frac{X^{3/2}}{80(\log X)^2}.$$

Given N, take $X = (N/9)^{1/3}$. All these n are at most N, and since $\log X \le \frac{1}{3}\log N$ there are at least $(N/9)^{1/2}\cdot 9/(80(\log N)^2) = \frac{3}{80}N^{1/2}/(\log N)^2$ of them. This proves the theorem with $c = 3/80$. Every step above that was stated for large X holds for $X \ge 6.3\cdot 10^7$ (the binding one is $L/(4\log X) - 3 - \log_2 L \ge L/(5\log X)$), so one may take $N_0 = 3\cdot 10^{24}$; we have not tried to optimise N_0. The last statement follows from Theorem 1.1. □

Proof of Corollary 1.3. For squarefree $n = pqr$ we have $P(n) = r$ and $n/P(n) = pq$. Theorem 1.1 gives the criterion, and the proof of Theorem 1.2 gives infinitely many n satisfying it. □

Remark 5.2. Lemma 5.1 uses nothing about primes except their number, and in a block with k primes it guarantees only about $k^2/2$ increasing triples. A larger count needs information on the distribution of primes inside the blocks. For example, cut each block I_t into ten sub-blocks of length $L/10$. Then any p, q, r taken from the sub-blocks numbered 0, 1 and 5 satisfy $r - q > q - p$. If almost all sub-blocks contained $\gg L/\log X$ primes, this would give $\gg N^{2/3}/(\log N)^3$ integers $n \le N$. Theorem 1.2 uses only Chebyshev-type information.

6 Computations

The programs are listed in the appendix. All of them examine the binomial coefficients themselves. None of them uses Lemmas 3.1–4.4.

Exhaustive check of Theorem 1.1. The program `verify_triples.c` runs through all prime triples $p < q < r$ with $3 \le p \le 30{,}000$ and $2(r-p)^2 < p$. For each $n = pqr$ and each of the three pairs $\{s, t\}$, it scans *all* multiples k of st up to $n/2$ and tests whether k is digit-dominated by n in both bases. By Lemma 2.2, Lemma 2.3 (the gcd is never 1) and the fact that $k = r$ always gives $\gcd(n, \binom{n}{r}) = pq$, while $q < r < pq$, this determines $f(n)$ exactly: $f = p$ if the pair $\{q, r\}$ admits such a k, otherwise $f = q$ if $\{p, r\}$ does, otherwise $f = r$ if $\{p, q\}$ does, and otherwise $f = pq$.

range	triples	increasing gaps	Theorem 1.1 confirmed
$3 \le p \le 30{,}000$, $2(r-p)^2 < p$	102,122	50,413	102,122

Independent check with Legendre's formula. The program `legendre_check.c` computes $v_s(\binom{n}{k}) = \sum_{i \ge 1}(\lfloor n/s^i \rfloor - \lfloor k/s^i \rfloor - \lfloor (n-k)/s^i \rfloor)$ for every $k \le n/2$ and every $s \in \{p, q, r\}$. It uses neither Lucas' theorem nor the reduction to multiples of st.

p	q	r	n	$f(n)$	comment
1009	1013	1019	1,041,537,223	1,022,117	(2), $r-q > q-p$: $f = pq$
1013	1019	1021	1,053,924,187	1,013	(2), $r-q \le q-p$: $f = p$
401	409	419	68,719,771	409	outside (2), $r-q > q-p$: $f = q$
3389	3391	3433	39,452,375,867	3,389	$1.75\,b^2 < p < 2b^2$, $r-q > q-p$: $f = p$

The first two rows are covered by Theorem 1.1: $f(n) = pq$ for $(1009, 1013, 1019)$, where $6 > 4$, and $f(n) = p$ for $(1013, 1019, 1021)$, where $2 \le 6$. The minimiser in the first row is $k = r$, as in the proof. The last two rows lie outside (2) and are discussed next.

Sharpness of (2). Replace the constant 2 in (2) by κ and rerun `verify_triples.c` for $p \le 8000$:

condition	triples	increasing gaps	formula of Theorem 1.1 holds
$2\,(r-p)^2 < p$, $p \le 8000$	10,089	4,947	10,089
$1.9\,(r-p)^2 < p$, $p \le 8000$	10,635	5,211	10,635
$1.75\,(r-p)^2 < p$, $p \le 8000$	11,563	5,668	11,556
$(r-p)^2 < p$, $p \le 8000$	21,324	10,508	18,195

For $\kappa = 1.75$ the formula fails for exactly seven triples. The smallest is $(3389, 3391, 3433)$: here $r - q = 42 > 2 = q - p$, but $f(n) = 3389 = p$, as the Legendre check confirms. Its pair $\{q, r\}$ admits a doubly dominated k, because $c(b+c) = 42 \cdot 86 = 3612 > r = 3433$. In fact the minimiser found by the Legendre check is $k = 442\,369\,514 = qr \cdot 38$, and $38 = \lceil (r - bc)/c \rceil$ is the value given by the formula for j^* in Lemma 4.4, although that lemma assumes $c \le a$ and does not apply here. Thus the constant 2 cannot be replaced by 1.75. Outside (2) the orientation alone does not decide: $(401, 409, 419)$ has $r - q = 10 > 8 = q - p$, but $f(n) = 409$.

The smallest examples. The smallest triples in the range (2) with increasing gaps are listed below. For each, Theorem 1.1 gives $f(n) = pq$, and `verify_triples.c` confirms it.

p	q	r	$n = pqr$	$f(n) = pq$
101	103	107	1 113 121	10 403
107	109	113	1 317 919	11 663
149	151	157	3 532 343	22 499
191	193	197	7 262 011	36 863
191	193	199	7 335 737	36 863
223	227	233	11 794 693	50 621

The function f for small n. The program `list_large_f.c` lists the composite n with $f(n) > n^{1/2}$. By Lemma 2.3 it only needs to examine k sharing a divisor $d \geq n^{1/2}$ with n, and it computes each $\gcd(n, \binom{n}{k})$ by Kummer's theorem. There are 793 such $n \leq 20{,}000$; the first 24 of them are 30, 70, 84, 105, 132, 140, 154, 165, 168, 182, 260, 273, 280, 306, 357, 364, 380, 385, 399, 418, 420, 455, 495, 520. For $n \leq 3000$ this list (137 values) agrees exactly with the output of `exhaust_f.c`, which examines every k. None of these n comes from Theorem 1.1: the smallest n it produces is $101 \cdot 103 \cdot 107 = 1\,113\,121$. So all the small examples have other shapes; Theorem 1.1 describes one infinite family, not all of them. Among composite $n \leq 12{,}000$ the largest value of $\log f(n)/\log n$ is 0.6815, attained at $n = 7293 = 3 \cdot 11 \cdot 13 \cdot 17$ with $f(n) = 429 = n/P(n)$.

Four nearby primes. One might hope that four nearby primes $p_1 < p_2 < p_3 < p_4$ give $f(n) = n/P(n) \approx n^{3/4}$. The program `cluster4.c` examined all 54 quadruples of primes with $20{,}000 \leq p_1 \leq 20{,}400$ and $p_4 - p_1 \leq 27$, so that $(p_4 - p_1)^3 < p_1$. In *every* case some $k \leq n/2$ is digit-dominated in two of the four bases, so $f(n) < n/P(n)$. This includes the six quadruples whose gaps strictly increase. For example, for $(20021, 20023, 20029, 20047)$ the multiplier $j = 1113$ of p_3p_4 works. With four primes the multiplier j has two base-p digits rather than one, and we do not know how to extend the argument of Section 4.

7 Concluding remarks

Theorem 1.2 answers Erdős Problem #700(b) affirmatively. Parts (a) and (c) remain open. Theorem 1.1 answers (a) completely for the products pqr in the range (2): $f(n) = n/P(n)$ exactly when the gaps increase, and $f(n) = p(n)$ otherwise. Two questions suggested by this work are:

1. What is $\limsup_n \log f(n)/\log n$? Theorem 1.2 shows that it is at least $2/3$. The value 0.6815 at $n = 7293$ shows that larger exponents occur for individual n; we do not know whether they occur infinitely often. A positive answer to (c) would not contradict a value of 1.

2. Is there an analogue of Theorem 1.1 for products of $m \geq 4$ primes in some other configuration? In the experiments of Section 6, clusters of four nearby primes did not work.

References

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A Programs

All programs are in C99 (with `unsigned __int128` where noted) and were compiled with `gcc -O2`; `legendre_check.c` also uses `-fopenmp`. Commands and outputs:

```
$ gcc -O2 -o verify_triples verify_triples.c
$ ./verify_triples 3 30000 2 | tail -1
p in [3,30000], kappa=2.000: triples=102122 (increasing gaps: 50413), formula holds for 102122
$ for K in 2 1.9 1.75 1; do ./verify_triples 3 8000 $K | tail -1; done
p in [3,8000], kappa=2.000: triples=10089 (increasing gaps: 4947), formula holds for 10089
p in [3,8000], kappa=1.900: triples=10635 (increasing gaps: 5211), formula holds for 10635
p in [3,8000], kappa=1.750: triples=11563 (increasing gaps: 5668), formula holds for 11556
p in [3,8000], kappa=1.000: triples=21324 (increasing gaps: 10508), formula holds for 18195
$ ./verify_triples 3 8000 1.75 | grep MISMATCH
MISMATCH p=3389 q=3391 r=3433 f=3389 formula=11492099
MISMATCH p=4091 q=4093 r=4139 f=4091 formula=16744463
MISMATCH p=5279 q=5281 r=5333 f=5279 formula=27878399
MISMATCH p=5501 q=5503 r=5557 f=5501 formula=30272003
MISMATCH p=6761 q=6763 r=6823 f=6761 formula=45724643
MISMATCH p=6779 q=6781 r=6841 f=6779 formula=45968399
MISMATCH p=7757 q=7759 r=7823 f=7757 formula=60186563
$ gcc -O2 -fopenmp -o legendre_check legendre_check.c
$ for t in "1009 1013 1019" "1013 1019 1021" "401 409 419" "3389 3391 3433"; do ./legendre_check $t; done
p=1009 q=1013 r=1019 n=1041537223: f(n)=1022117 (first minimiser k=1019); pq=1022117; f(n)^2>n: yes
p=1013 q=1019 r=1021 n=1053924187: f(n)=1013 (first minimiser k=523320697); pq=1032247; f(n)^2>n: no
p=401 q=409 r=419 n=68719771: f(n)=409 (first minimiser k=3864437); pq=164009; f(n)^2>n: no
p=3389 q=3391 r=3433 n=39452375867: f(n)=3389 (first minimiser k=442369514); pq=11492099; f(n)^2>n: no
$ gcc -O2 -o list_large_f list_large_f.c && ./list_large_f 20000 > list20000.txt
composite n <= 20000 with f(n) > sqrt(n): 793
$ gcc -O2 -o exhaust_f exhaust_f.c
$ ./exhaust_f 3000 | awk '$2*$2>$1{print $1}' > a1.txt; awk '$1<=3000' list20000.txt > a2.txt; cmp a1.txt
    a2.txt && wc -l < a1.txt
137
$ ./exhaust_f 12000 | awk '{printf "%d %d %.4f\n",$1,$2,log($2)/log($1)}' | sort -k3 -nr | head -1
7293 429 0.6815
$ gcc -O2 -o cluster4 cluster4.c && ./cluster4 20000 20400 27 | tail -1
clusters=54 without double domination=0
```

A.1 `verify_triples.c`

```
/* verify_triples.c -- exhaustive check of Theorem 1.1.
   For all prime triples p<q<r with LO <= p <= HI and KAPPA*(r-p)^2 < p, compute f(pqr) exactly and
   compare it with the formula  f = pq if r-q > q-p,  f = p otherwise.
   Exact computation: n=pqr is squarefree, so gcd(n,C(n,k)) is the product of the primes s in {p,q,r}
   with s | C(n,k) (Lemma 2.2).  A value smaller than pq occurs iff some k <= n/2 has two primes s,t
   with s,t not dividing C(n,k); by Lucas' theorem this means k = s*t*j is digit-dominated by n in
   bases s and t.  We scan ALL multiples k of s*t up to n/2 (a fast last-digit prefilter only skips
```

```
   k that fail domination in its last digit), for each of the three pairs.
   Then f = p if {q,r} admits such k, else q if {p,r} does, else r if {p,q} does, else pq.
   Usage: verify_triples LO HI KAPPA      (KAPPA=2 is the hypothesis of Theorem 1.1) */
#include <stdio.h>
#include <stdlib.h>
typedef unsigned long long u64;
static int dom(u64 k, u64 n, u64 s){ while(k>0){ if(k%s > n%s) return 0; k/=s; n/=s; } return 1; }
static int pair_dominated(u64 n, u64 s, u64 t){           /* exists k<=n/2, s*t|k, k dom_s n and k dom_t n
      */
  u64 st=s*t, ls=(n/s)%s, lt=(n/t)%t;                     /* last digits of n/s (base s), n/t (base t) */
  for(u64 k=st; k<=n/2; k+=st){ u64 j=k/st;
    if(((t%s)*(j%s))%s > ls) continue;                    /* last base-s digit of k/s = t*j */
    if(((s%t)*(j%t))%t > lt) continue;                    /* last base-t digit of k/t = s*j */
    if(dom(k,n,s) && dom(k,n,t)) return 1; }
  return 0; }
int main(int argc, char **argv){
  u64 lo=strtoull(argv[1],0,10), hi=strtoull(argv[2],0,10); double kappa=atof(argv[3]);
  u64 M=hi+5000; char *comp=calloc(M+1,1);
  for(u64 i=2;i*i<=M;i++) if(!comp[i]) for(u64 j=i*i;j<=M;j+=i) comp[j]=1;
  long tot=0, ok=0, incr=0;
  for(u64 p=(lo<3?3:lo); p<=hi; p++){ if(comp[p]) continue;
    for(u64 q=p+2; kappa*(double)(q-p)*(double)(q-p) < (double)p; q++){ if(comp[q]) continue;
      for(u64 r=q+2; kappa*(double)(r-p)*(double)(r-p) < (double)p; r++){ if(comp[r]) continue;
        u64 n=p*q*r, f;
        if(pair_dominated(n,q,r)) f=p; else if(pair_dominated(n,p,r)) f=q;
        else if(pair_dominated(n,p,q)) f=r; else f=p*q;
        u64 formula = (r-q > q-p) ? p*q : p;
        tot++; incr += (r-q > q-p); if(f==formula) ok++;
        else printf("MISMATCH p=%llu q=%llu r=%llu f=%llu formula=%llu\n",p,q,r,f,formula);
      } } }
  printf("p in [%llu,%llu], kappa=%.3f: triples=%ld (increasing gaps: %ld), formula holds for %ld\n",
         lo,hi,kappa,tot,incr,ok);
  return 0; }
```

A.2 legendre_check.c

```
/* legendre_check.c -- independent check of single values f(pqr), with no use of Lucas' theorem.
   f(n) = min over 1<k<=n/2 of gcd(n,C(n,k)); for squarefree n = pqr the gcd is the product of the
   primes s with v_s(C(n,k)) >= 1, and v_s(C(n,k)) is computed by Legendre's formula
   sum_i ( floor(n/s^i) - floor(k/s^i) - floor((n-k)/s^i) ).  Every k is examined.
   Usage: legendre_check p q r        (compile with -fopenmp) */
#include <stdio.h>
#include <stdlib.h>
typedef unsigned long long u64;
static int vleg(u64 n, u64 k, u64 s){ int v=0; u64 m=s;
  for(;;){ v += (int)(n/m - k/m - (n-k)/m); if(m > n/s) break; m*=s; } return v; }
int main(int argc, char **argv){
  u64 p=strtoull(argv[1],0,10), q=strtoull(argv[2],0,10), r=strtoull(argv[3],0,10), n=p*q*r;
  u64 best=n, bestk=0;
  #pragma omp parallel
  { u64 lb=n, lk=0;
    #pragma omp for schedule(static)
    for(u64 k=2;k<=n/2;k++){ u64 g=1;
      if(vleg(n,k,p)) g*=p; if(vleg(n,k,q)) g*=q; if(vleg(n,k,r)) g*=r;
      if(g<lb){ lb=g; lk=k; } }
    #pragma omp critical
    { if(lb<best || (lb==best && lk<bestk)){ best=lb; bestk=lk; } } }
  printf("p=%llu q=%llu r=%llu n=%llu: f(n)=%llu (first minimiser k=%llu); pq=%llu; f(n)^2>n: %s\n",
         p,q,r,n,best,bestk,p*q,(best*best>n)?"yes":"no");
  return 0; }
```

A.3 list_large_f.c

```
/* list_large_f.c -- all composite n <= N with f(n) > sqrt(n).
   By Lemma 2.3, gcd(n,C(n,k)) >= n/gcd(n,k); so only k with gcd(n,k) >= sqrt(n), i.e. k a multiple of a
   divisor d >= sqrt(n) of n, can give gcd(n,C(n,k)) <= sqrt(n).  For those k the gcd is computed exactly
   with Kummer's theorem: v_s(C(n,k)) = number of carries when adding k and n-k in base s.
   Usage: list_large_f N */
```

```
#include <stdio.h>
#include <stdlib.h>
typedef unsigned long long u64; typedef unsigned __int128 u128;
static int carries(u64 x, u64 y, u64 s, int cap){ int c=0, cy=0;
  while(x>0 || y>0 || cy){ u64 t=x%s+y%s+cy; cy=(t>=s); c+=cy; if(c>=cap) return cap; x/=s; y/=s; } return
      c; }
int main(int argc, char **argv){
  u64 N=strtoull(argv[1],0,10); long count=0;
  for(u64 n=4;n<=N;n++){
    u64 ps[20]; int es[20], w=0; u64 t=n;
    for(u64 s=2;s*s<=t;s++) if(t%s==0){ int e=0; while(t%s==0){ t/=s; e++; } ps[w]=s; es[w]=e; w++; }
    if(t>1){ ps[w]=t; es[w]=1; w++; }
    if(w==1 && es[0]==1) continue;                          /* prime */
    u64 divs[4096]; int nd=1; divs[0]=1;
    for(int i=0;i<w;i++){ int cur=nd; u64 m=1; for(int e=1;e<=es[i];e++){ m*=ps[i]; for(int a=0;a<cur;a++)
      divs[nd++]=divs[a]*m; } }
    int good=1;
    for(int a=0;a<nd && good;a++){ u64 d=divs[a]; if((u128)d*d < n || d > n/2) continue;
      for(u64 k=d;k<=n/2 && good;k+=d){ u128 g=1;
        for(int i=0;i<w;i++){ int c=carries(k,n-k,ps[i],es[i]); for(int e=0;e<c;e++) g*=ps[i]; }
        if(g*g <= n) good=0; } }
    if(good){ count++; printf("%llu\n",n); } }
  fprintf(stderr,"composite n <= %llu with f(n) > sqrt(n): %ld\n",N,count);
  return 0; }
```

A.4 exhaust_f.c

```
/* exhaust_f.c -- f(n) for every composite n <= N by examining every k (Kummer's theorem).
   Output per line: n f(n) first-minimiser-k n/P(n) n/Q(n) omega(n).   Usage: exhaust_f N */
#include <stdio.h>
#include <stdlib.h>
static int carries(long x, long y, long s, int cap){ int c=0, cy=0;
  while(x>0 || y>0 || cy){ long t=x%s+y%s+cy; cy=(t>=s); c+=cy; if(c>=cap) return cap; x/=s; y/=s; }
     return c; }
int main(int argc, char **argv){
  long N=atol(argv[1]);
  for(long n=4;n<=N;n++){
    long ps[16]; int es[16], w=0; long t=n;
    for(long s=2;s*s<=t;s++) if(t%s==0){ int e=0; while(t%s==0){ t/=s; e++; } ps[w]=s; es[w]=e; w++; }
    if(t>1){ ps[w]=t; es[w]=1; w++; }
    if(w==1 && es[0]==1) continue;
    long best=n, bk=0;
    for(long k=2;k<=n/2;k++){ long g=1;
      for(int i=0;i<w && g<best;i++){ int c=carries(k,n-k,ps[i],es[i]); for(int e=0;e<c;e++) g*=ps[i]; }
      if(g<best){ best=g; bk=k; } }
    long Q=1; for(int i=0;i<w;i++){ long m=1; for(int e=0;e<es[i];e++) m*=ps[i]; if(m>Q) Q=m; }
    printf("%ld %ld %ld %ld %ld %d\n", n, best, bk, n/ps[w-1], n/Q, w); }
  return 0; }
```

A.5 cluster4.c

```
/* cluster4.c -- four nearby primes p1<p2<p3<p4 with p4-p1 <= H and p1 in [LO,HI], n = p1p2p3p4.
   Reports whether some k <= n/2 is digit-dominated by n in two of the four bases (Lucas), i.e. whether
   two of the primes fail to divide C(n,k).  If so, f(n) < n/P(n).   Usage: cluster4 LO HI H */
#include <stdio.h>
#include <stdlib.h>
typedef unsigned __int128 u128; typedef unsigned long long u64;
static int dom(u128 k, u128 n, u64 s){ while(k>0){ if(k%s > n%s) return 0; k/=s; n/=s; } return 1; }
static int isprime(u64 x){ if(x<2) return 0; for(u64 d=2; d*d<=x; d++) if(x%d==0) return 0; return 1; }
int main(int argc, char **argv){
  u64 lo=strtoull(argv[1],0,10), hi=strtoull(argv[2],0,10), H=strtoull(argv[3],0,10);
  u64 pr[200000]; int np=0; for(u64 x=lo;x<=hi+H;x++) if(isprime(x)) pr[np++]=x;
  long tot=0, good=0;
  for(int a=0;a<np && pr[a]<=hi;a++) for(int b=a+1;b<np && pr[b]-pr[a]<=H;b++)
  for(int c=b+1;c<np && pr[c]-pr[a]<=H;c++) for(int d=c+1;d<np && pr[d]-pr[a]<=H;d++){
    u64 P[4]={pr[a],pr[b],pr[c],pr[d]}; u128 n=(u128)P[0]*P[1]*P[2]*P[3]; int ok=1, bi=-1, bj=-1; u64 jw
     =0;
    for(int i=0;i<4 && ok;i++) for(int i2=i+1;i2<4 && ok;i2++){ u64 s=P[i], t=P[i2]; u128 st=(u128)s*t, J=
     n/(2*st);
```

```
    u64 ls=(u64)((n/s)%s), lt=(u64)((n/t)%t);
    for(u128 j=1;j<=J;j++){ if((u64)(((u128)(t%s)*(j%s))%s) > ls) continue; if((u64)(((u128)(s%t)*(j%t))
  %t) > lt) continue;
      if(dom(st*j,n,s) && dom(st*j,n,t)){ ok=0; bi=i; bj=i2; jw=(u64)j; break; } } }
  tot++; good+=ok;
  printf("%llu %llu %llu %llu gaps %llu %llu %llu : %s", P[0],P[1],P[2],P[3],P[1]-P[0],P[2]-P[1],P[3]-P
   [2], ok?"no double domination":"double domination");
  if(!ok) printf(" (pair %d,%d, k = p%d*p%d*%llu)", bi+1, bj+1, bi+1, bj+1, jw); printf("\n"); fflush(
   stdout); }
printf("clusters=%ld without double domination=%ld\n", tot, good);
return 0; }
```